

Network Measurement SDK - Integration Guide

This guide describes how to integrate and use the Network Measurement SDK in your Android application.

1. Project Configuration

Ensure the SDK is added to your project dependencies.

```
// build.gradle.kts (Module)
dependencies {

    implementation("com.github.primetechsoltutions:network-measurement-unified-sdk:X.X.X") // Replace with SDK Version
}
```

```
//settings.gradle.kts (Project Level)
maven {
    url = uri("https://jitpack.io")
}
```

2. Initialization

Initialize the SDK in your Activity before calling any capture methods.

```
val networkUploader = NetworkDataUploader()
networkUploader.init(this, "Your Application Name")
```

3. Capturing Network Data

Parameter	Type	Requirement	Description
msisdn	String	Required	User mobile number.
integratedAppVersion	String	Required	Host app version.

Parameter	Type	Requirement	Description
sdkInitiateTimeStamp	String	Required	Format: yyyy-MM-dd'T'HH:mm:ss.
integratedAppEventName	String	Required	Unique name for the event.
uploadType	UploadType	Required	NetworkDataCapture or FTPNetworkDataCapture.
userLatitude	Double	<i>Optional</i>	Fallback latitude.
userLongitude	Double	<i>Optional</i>	Fallback longitude.

startUploading() Parameters

Standard Capture

```
networkUploader.startUploading(
    msisdn = "01XXXXXXXXXX",
    integratedAppVersion = "1.0.0",
    sdkInitiateTimeStamp = "2024-02-27T22:15:00", // yyyy-MM-dd'T'HH:mm:ss
    integratedAppEventName = "Home_Page_Visit",
    uploadType = UploadType.NetworkDataCapture
) { success, status ->
    if (success) Log.d("SDK", "Data capture scheduled")
}
```

FTP Diagnostic Capture

```
networkUploader.startUploading(
    msisdn = "01XXXXXXXXXX",
    integratedAppVersion = "1.0.0",
    sdkInitiateTimeStamp = "2024-02-27T22:15:00",
    integratedAppEventName = "Manual_Diagnostic",
    uploadType = UploadType.FTPNetworkDataCapture
) { success, status ->
    // status.message contains the JSON response
    Log.d("SDK", "FTP Result: ${status.message}")
}
```

4. Understanding Responses

FTP Response Examples

Success/Pass Response

The status.message returns a JSON string containing the diagnostic values:

```
{
  "status": "Success",
  "testResult": "Pass",
  "statusCode": 200,
  "message": "Your network assessment was successful.",
  "data": {

    "assessmentId": 119,
    "networkData": { "RSRP": -105, "SNR": 18, "RSRQ": -3 },
    "cellInfo": {
      "cellName": "DHK_K1469_1",
      "eNodeBName": "DHK_K1469",
      "nbhDIThroughputMbps": 11.5,
      "nbhTrafficGB": 0.5
    },
    "speedPair": { "ulSpeedKbps": 5000, "dlSpeedKbps": 5000 },
    "userInfo": {
      "msisdn": "01XXXXXXXXXX",
      "deviceManufacture": "OnePlus",
      "deviceModel": "ONEPLUSA6000",
      "deviceOsVersion": "30",
      "latitude": 23.7515,
      "longitude": 90.3933
    }
  }
}
```

Success/Failed Response

The status.message returns a JSON string containing the diagnostic values:

```
{
  "status": "Success",
  "testResult": "Failed",
  "statusCode": 200,
  "message": "Your network assessment was successful.",
  "data": {
    "assessmentId": 120,
    "networkData": { "RSRP": -120, "SNR": 18, "RSRQ": -3 },
    "cellInfo": {
      "cellName": "DHK_K1469_1",
      "eNodeBName": "DHK_K1469",
      "nbhDlThroughputMbps": 11.5,
      "nbhTrafficGB": 0.5
    },
    "speedPair": { "ulSpeedKbps": 6763, "dlSpeedKbps": 3487 },
    "userInfo": {
      "msisdn": "01XXXXXXXXXX",
      "deviceManufacture": "OnePlus",
      "deviceModel": "ONEPLUSA6000",
      "deviceOsVersion": "30",
      "latitude": 23.7515,
      "longitude": 90.3933
    }
  }
}
```

Failed Response (Example: Permissions Missing)

```
{
  "status": "Failed",
  "testResult": "Failed",
  "statusCode": 400,
  "message": "Permission are required for FTP Capture",
  "data": null
}
```

Failed Response (Example: Timeout)

```
{
  "status": "Failed",
  "testResult": "Failed",
  "statusCode": 408,
  "message": "Network assessment timed out. Please check your internet connection.",
  "data": null
}
```

Failed Response (Example:Non Banglalink 4G)

```
{
  "status": "Failed",
  "testResult": "Failed",
  "statusCode": 400,
  "message": "FTP Capture is only supported on Banglalink 4G (LTE) technology."
}
```

Failed Response (Example:Non Mobile Network)

```
{
  "status": "Failed",
  "testResult": "Failed",
  "statusCode": 400,
  "message": "FTP Capture requires mobile data. Please disable Wi-Fi and ensure Banglalink 4G is active."
}
```

5. Requirements & Constraints

Operational Constraints

- **Network:** Requires an active connection (`NetworkType.CONNECTED`).
- **Battery:** Requires the device battery to **not be low**.

Required Permissions

- **INTERNET, ACCESS_FINE_LOCATION, ACCESS_COARSE_LOCATION, READ_PHONE_STATE.**

[!IMPORTANT] Without these permissions, the app may not be able to collect accurate data or function correctly.

Platform Support

This SDK is exclusively supported on the **Android Platform**.

6. Best Practices

[!WARNING] **Production Readiness:** Perform cross-testing with the SDK before releasing any app update to production to verify end-to-end functionality and data consistency.

About the Network Measurement SDK The Network Measurement SDK (Android) is a comprehensive tool for real-time mobile network analysis. It provides integrated signal telemetry, throughput measurement, and diagnostic reporting to help host applications maintain optimal connectivity for their users.